PCBC H Parametrics Practice Quiz Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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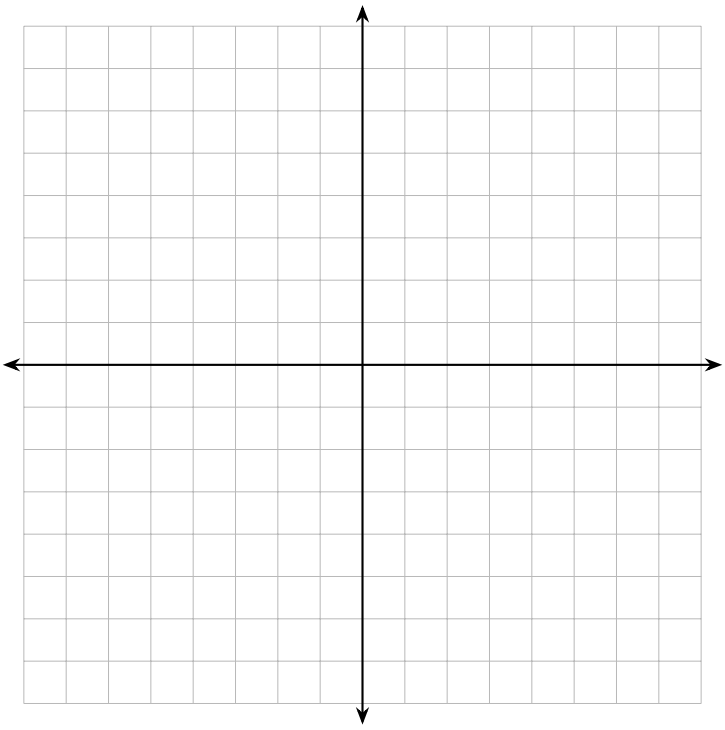
Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

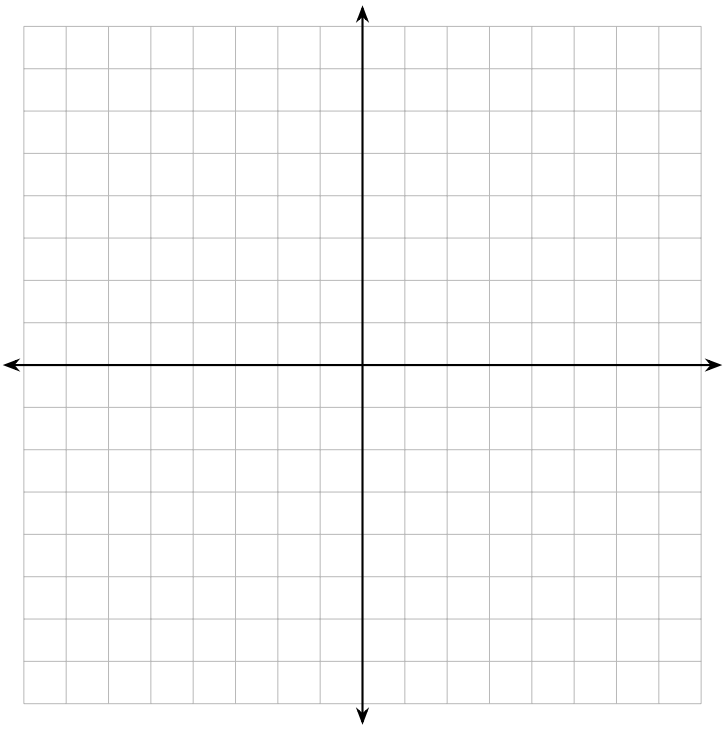
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = \sqrt{t}, \quad y = 2t - 3, \quad \text{for } t \text{ in } [0,4]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 2\tan t, \quad y = 3\sec t, \quad \text{for } t \text{ in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 + 6x + 5$ **3 pts**

Representation 1 *Representation 2*

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Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations **3 pts**

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

A football is punted from a height of 2.5 ft at an angle of inclination of 40° with a velocity of 80 ft/sec. Write the parametric equations that model its path. For how many seconds is the football in flight? What horizontal distance does it travel?